Integral Form Of Dottie's Number Via Contour Integration

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The constant known as Dottie's Number, here denoted by Φ, is the real solution to the transcendental equation $cos(x) = x$. By evaluating the inverse of $cos(x) - x$ at 0 through methods of complex analysis, an integral form of the constant is attained.

1 Setup

Consider expression (1) shown below: when integrated over a contour containing the root of $f(z) - w$, it equates to $f^{-1}(w)$. This can be seen either from the substitution $u = f(z)$, leaving the Cauchy's Theorem form of $\frac{1}{2\pi i} \int \frac{f^{-1}(u)}{u-w} = f^{-1}(w)$ behind, or by computing its residue at $f^{-1}(w)$ with L'Hôpital.

$$f^{-1}(w) = \frac{1}{2\pi i} \oint \frac{z f'(z)}{f(z) - w} dz \tag{1}$$

Thus, function inversion can be achieved for any holomorphic function f on an appropriate domain.

For the purposes of solving an equation such as $\cos(x) = x$, we only need to know a lower and upper bound for the solution since, by the Residue Theorem, any path containing the region would equate to the sum of its solutions.

With these in mind, our solution Φ becomes:

$$-2\pi i \Phi = \oint \frac{z(\sin(z) + 1)}{\cos(z) - z} dz$$

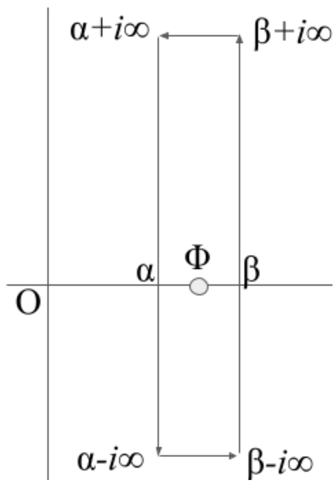
For notation, define $F(z) = \frac{z(\sin z + 1)}{\cos z - z}$. Additionally, note the useful property that

$$\lim_{y\to\infty} \frac{\sin(x+iy)+1}{\cos(x+iy) - x - iy} = \lim_{y\to\infty} \frac{\sin x + i\cos x \tanh y + \frac{1}{\cosh y}}{\cos x - i \sin x \tanh y - \frac{x}{\cosh y} - \frac{iy}{\cosh y}} = i$$

Therefore,

$$\lim_{y\to\infty} \Im F(x+iy) = x \tag{2}$$

(Where $\Im z$ is the imaginary part of z) This simplification, along with the fact that $F(\overline{z}) = \overline{F(z)}$, motivates the use of an infinitely tall, rectangular contour surrounding the value of Φ (figure below). $0 < \Phi \approx 0.739 < \frac{\pi}{2}$.



2 Integration

$$-2\pi i\Phi = \int_{-\infty}^{\infty} F\left(\beta + it\right) idt - \int_{-\infty}^{\infty} F\left(\alpha + it\right) idt + \int_{\alpha}^{\beta} F\left(t - i\infty\right) dt - \int_{\alpha}^{\beta} F\left(t + i\infty\right) dt$$

By the conjugate symmetry of F and (2) these last two integrals reduce to:

$$2i\int_{\alpha}^{\beta} \Im F\left(t - i\infty\right) dt = 2i\int_{\alpha}^{\beta} tdt = i\left(\beta^2 - \alpha^2\right)$$

To deal with the remaining integrals, we select α and β to provide the greatest amount of cancellation; I have let $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ play this role, respectively.

$$\Phi = \frac{-1}{2\pi}\int_{-\infty}^{\infty} \frac{\left(\frac{\pi}{2} + it\right)\left(\sin\left(\frac{\pi}{2} + it\right) + 1\right)}{\cos\left(\frac{\pi}{2} + it\right) - \left(\frac{\pi}{2} + it\right)} - \frac{\left(\frac{-\pi}{2} + it\right)\left(\sin\left(\frac{-\pi}{2} + it\right) + 1\right)}{\cos\left(\frac{-\pi}{2} + it\right) - \left(\frac{-\pi}{2} + it\right)} dt \tag{3}$$

$$= \frac{-1}{2\pi}\int_{0}^{\infty} \left(\frac{-2t\left(t + \sinh\left(t\right)\right)\left(1 + \cosh\left(t\right)\right)}{\left(t + \sinh\left(t\right)\right)^2 + \frac{\pi^2}{4}} + \frac{2t\left(t - \sinh t\right)\left(1 - \cosh\left(t\right)\right)}{\left(t - \sinh\left(t\right)\right)^2 + \frac{\pi^2}{4}}\right) dt - \frac{\pi}{2}$$

Integrating by parts on each term and combining with addition of logs, one obtains;

$$\Phi = \frac{\pi}{2} - \frac{1}{2\pi}\int_{0}^{\infty} \ln\left(\frac{\left(t + \sinh\left(t\right)\right)^2 + \frac{\pi^2}{4}}{\left(t - \sinh\left(t\right)\right)^2 + \frac{\pi^2}{4}}\right) dt \tag{4}$$

Alternative choices for α and β and rearrangements can give:

$$\Phi = \frac{\pi}{2} - \frac{1}{2\pi}\int_{0}^{\infty} \ln\left(1 + \frac{4t\sinh\left(t\right)}{\left(t - \sinh\left(t\right)\right)^2 + \frac{\pi^2}{4}}\right) dt$$

Or:

$$\Phi = \frac{3\pi}{8} - \frac{1}{2\pi}\int_{0}^{\infty} \ln\left(1 + \frac{2t\sinh\left(t\right) + \frac{\pi^2}{4} - 1}{t^2 + \cosh\left(t\right)^2}\right) dt$$

Similar integration by parts tacits produces perhaps the simpliest of these:

$$\boxed{\Phi = \frac{\pi}{2} - \frac{1}{2\pi}\int_{0}^{\infty} \ln\left(1 + \pi\frac{2\cosh\left(t\right) + \pi}{t^2 + \cosh\left(t\right)^2}\right) dt}$$

As a side note, the similarlities between these integrals can be exploited to produce the solution for the following integral:

$$-\frac{\pi^2}{4} = \int_{0}^{\infty} \ln\left(\frac{\left(t - \sinh\left(t\right)\right)^2 + \frac{\pi^2}{4}}{t^2 + \cosh\left(t\right)^2}\right) dt$$

3 Conclusions

Taking (1) under closer scrutiny, we can slightly generalize the expression to return not the inverse, but rather an arbitrary function evaluated at the inverse of f. This expression is identical to the generalized argument theorem for any $f(z)$ without poles.

$$h\left(f^{-1}(w)\right) = \frac{1}{2\pi i}\oint \frac{h(z)\, f'(z)}{f(z)-w}dz$$

Further, the fact that the $F(z)$ constructed in Setup is conjugate symmetric $(F(\overline{z}) = \overline{F(z)})$ is no coincidence due to the Schwarz Reflection Principle which applies to all anaylitc functions that are real-valued on the real-axis. It is this universal conjugate symmetry which motivates the general form for a root-finding contour integral along real-axis symmetrical **rectangular** boundaries (of height h):

Let $f(x)$ be real-valued on the real-axis such that the sole root, ψ, to $f(x) - w$, for real w, in the domain $\alpha < \Re\, z < \beta$ lies on the real-axis.

$$2\pi i\psi = \oint F(z)\, dz = 2\Re \int_0^h F(\beta + it)\, idt - 2\Re \int_0^h F(\alpha + it)\, idt + 2i\Im \int_\alpha^\beta F(t - ih)$$

Integration by parts on each of the $F(z) = z\frac{f'(z)}{f(z)-w}$ terms produces *shared* boundary terms at the corners of the rectangle, of which only those with an imaginary component remain.

$$\pi\psi = \int_0^h \ln\left|\frac{f(\alpha + it) - w}{f(\beta + it) - w}\right| + \Im \int_\alpha^\beta \ln\left(f(t + ih) - w\right) dt + \tag{5}$$

$$2\Im\left[(\alpha + ih)\ln\left(f(\alpha + ih) - w\right) - (\beta + ih)\ln\left(f(\beta + ih) - w\right)\right]$$